ZOO102 Final Term Notes

Regards By Admins:

VU Friends

Q.1 Write Electron Transport Chain and its location, purpose and also its components?

Electron Transport Chain:

It is an organized set of electron carriers and enzymes with enhanced oxidation reduction reactions which extract the energy of electrons.

Location of Electron Transport Chain:

Inner mitochondrial membrane

Purpose of Electron Transport Chain:

- Accept electrons from NADH and $FADH_2$
- Pass electrons through redox center
- Take the energy of electrons
- Move H^+ across inner mitochondrial membrane

Components of Electron Transport Chain:

- NADH dehydrogenase
- Flavoproteins: FAD, FMN
- Ubiquinone: CoE Q
- Cytochromes
- Fe-S proteins

Q.2 Explain ATP Synthesis….?

H^+ Gradient :

- Electron transport chain pumps H^+ from mitochondria to inter membrane space
- H^+ concentration is higher in inter membrane space than mitochondrial matrix

ATP Synthetase:

- Inner mitochondrial membrane is selectively permeable
- H^+ can come back to matrix through a membrane protein "ATP Synthetase

Mechanism of ATP Synthesis:

- H^+ come back to matrix through ATP synthetase
- Potential energy of H^+ is used for synthesis of ATP

Energy Yield in Oxidation of Glucose

1.ATP Yield During Anaerobic Respiration:

- In the absence of oxygen
- Glucose is not completely oxidized
- Anaerobic respiration= Glycolysis+ fermentation
- Energy comes from glycolysis only

2. ATP Yield During Aerobic Respiration :

- In the presence of oxygen
- Glucose is completely oxidized into CO_2 and H_2O
- Aerobic respiration= Glycolysis+ Krebs cycle+ electron transport chain

Q.3 Write Metabolism of Fats its imp.Digestion and storage of fats?

1. Importance of Fats :

- Important part of our diet
- Structure: membranes
- Important source of energy to our body
- More than double amount of energy as compared to carbohydrates and proteins
- 30% recommended
- Liver, heart and skeletal muscles meet half of energy requirement from fats
- Sole source of energy in hibernating animals
- Fats are actually triacylglycerides

2. Digestion of fats:

- Digestion occur in small intestine
- Bile…..Emulsification
- Lipase…Digested completed
- Absorb in small intestine and converted into triglycerides again

3. Storage of Fats:

Chylomicrons: Fats are transported in body after packing with cholesterol and proteins to form chylomicron.Stored in adipose tissues and skeletal muscles

4. Metabolism of Glycerol :

When required fats are hydrolyzed by enzymes into glycerol and fatty acids.Glycerol is converted to glyceraldehyde-3-phosphate .Enters into glycolysis

Q.4 Write Metabolism of Fatty Acids. and Why Fats Give More Energy?

Metabolism of Fatty Acids:

- ☐ In cell fatty acids oxidation mainly occurs in: Mitochondria
- ☐ Some amount: Endoplasmic reticulum , Peroxisome
- ☐ Breakdown of fatty acids yield: Acetyl CoA, NADH and $FADH_2$
- ☐ Acetyl CoA joins Krebs cycle

Acetyl CoA in Krebs cycle forms NADH and $FADH_2$

ADH and $FADH_2$ enter into electron transport chain for ATP production

Why Fats Give More Energy?

1g of fats give 2.5 times more energy than 1g of carbohydrates and protein.This is because they contain much higher amount of C-H bonds

Q.5 Metabolism of Proteins: Write imp. of protein and its digestion?

1. Importance of proteins:

- Muscle Fibers
- Bones
- Enzymes
- Hormones
- Antibodies
- Transporters
- Lens proteins of eyes

- Hair, nail, feather
- Spider web

2. Digestion of Proteins:

- In Stomach and small intestine
- Proteins are finally broken down to amino acids
- In small intestine amino acids are absorbed
- In Stomach and small intestine
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3. Break down of Amino Acids:

- Amino acids pool
- Excessive amino acids are degraded mainly in liver and skeletal muscles
- Also during starvation
- Amino acid converted to Keto acid after deamination

Q.6 Write Energy from Proteins and Excretion of Nitrogenous Wastes?

1. Energy from Proteins:

- Keto acids are funneled to Krebs cycle to yield ATP
- 1g of proteins give approximately 4kcal energy
- Approximately equal to glucose

2. Excretion of Nitrogenous Wastes:

Ammonia is formed during deamination of Amino acid.In aquatic organisms ammonia is released from moist surfaces.For release of ammonia large amount of water is required.In land animals ammonia is converted to urea .Urea excreted through urinary system.Moderate amount of water for urea.Animals with acute shortage of water, release uric acid.

Q.7 How we can control Metabolism?

1. Cells are Very Efficient:

Cells do not waste energy.If some molecule in excess, its synthesis is switched off

2. Feedback Inhibition:

The most common method for control of metabolism.In this method the product inhibits enzyme that catalyzes the key step.

e.g., ATP inhibits phosphofructokinase in glycolysis

3. Control of Catabolism:

If a cell is working hard its ATP is decreases.Decrease in ATP start anaerobic respiration.Anaerobic respiration slows when ATP is sufficient

4. Control of Anabolism:

In anabolism cells expends energy for biomolecules synthesis.Anabolism is mainly controlled by regulating the enzyme activity at strategic points

Q.8 Write effect of ATP,ADP and AMP and citric acid on metabolism?

Effect of ATP, ADP and AMP:

When high ATP level.Enzyme inhibited; resulting in inhibition of glycolysis.If high ADP or AMP level.Enzyme activated glycolysis starts

Effect of Citric Acid:

Citric acids produced in Krebs cycle

- citric acid in cytoplasm inhibits phosphofructokinase
- At low concentration of citric acid glycolysis accelerates

Example of Control:

- Phosphofructokinase
- By controlling the activity of this enzyme cell regulates its entire metabolic process
- Receptor sites for ATP, ADP, AMP, Citric acids

Q.9 What is Metabolic pool and write its important?

Metabolic pool:

- All the substances of metabolisms constitute metabolic pool
- It is actually reservoir of molecules upon which enzymes act.

Importance of Metabolic Pool:

All catabolic and anabolic reaction in cell use metabolic pool.

Substrates Products Reaction intermediates

Open Systems:

- In which compound may enter or leave the biochemical pathway at various steps
- Almost all the biochemical pathways are open system
- Metabolic intermediates of Glycolysis and Krebs cycle may withdrawn or put into by other metabolic pathways.

Q.10 Define reproduction and how many types of reproduction?

Defining Reproduction:

The process of producing new individuals. It is a the basic feature of all forms of life

Why There is Need for Reproduction?

With the passage of time animals get aged, new animals should take their place

It is essential for existence of species.Reproduction is essential for continuity of life on earth

Reproductive Potential:

- Capability to reproduce.
- All the animals have great reproductive potential
- if resource are unlimited a single species of fishes may fill the all oceans.

Types of Reproduction:

Animals reproduce by two basic means

(A).. Asexual reproduction

(B).. Sexual reproduction

Q.11 Explain ASexual Reproduction?

Define:

It is the type of reproduction in which offspring are produced without fertilization of gametes.

Identical Organisms:

- Basic method of cell division is mitosis
- No genetic recombination
- So the animals produced are identical to their parents and siblings

Method of Early Life on Earth:

- Early organisms reproduced by asexual method
- This was the only method of reproduction in first 2 billion years of life
- During this life evolved slowly because of less genetic diversity

Animals With Asexual Reproduction:

- It is common among:
- Protozoa; Amoeba, paramecium, Euglena
- Lower invertebrates; hydra, jellyfish, flatworms
- Asexual reproduction is correlated with ability to regenerate

Types of Asexual Reproduction in Animals:

1) Fission

2) Budding

3) Fragmentation

4) Parthenogenesis

Q.12 Explain Methods of Asexual Reproduction in fission?

Fission:

L. *fissio, the act of splitting* .It is type of asexual reproduction in which one organisms splits into two or more organisms.The organisms divides by inward furrowing

Types of Fission:

There are two basic types of fission

(i) Binary fission
(ii) Multiple fission

(i) Binary Fission:

- It is the process of fission in which one organism divided into two organisms.
- Both are approximately equal size
- Binary fission may be:
- Asymmetrical
- Transverse
- . Binary Fission
- Asymmetrical binary fission
- Longitudinal

(i) Multiple Fission:

- It is a number of successive cell divisions of nucleus followed by cytoplasm in which one cell divides into many cells
- E.g., amoeba, tapeworm

Q.13 Write Methods of Asexual Reproduction in Budding?

Budding:

- Asexual reproduction in which an outgrowth appears on the body of animal which grows into complete organism
- Hydra, Obelia, sponges

A group of cells rapidly divides and form a small outgrowth called bud. This bud increases rapidly and form a complete organism

Colony:

- Sometimes the buds remain attached with the parental organism and form a closely associated group of individuals called a colony.Members of a colony belong to same species

Gemmules:

In sponges internal buds are produces.The internal buds are called as gemmules.Gemmule is a collection of cells surrounded by body wall.When parent die each gemmule gives rise to a new individual**.**

Q.14 Write Methods of Asexual Reproduction in Fragmentation?

Fragmentation:

It is the process of asexual reproduction in which an animal breaks into two or more pieces and each piece is capable of growing into a new organism

Fragmentation in Different Groups of Animals:

Fragmentation is common in different groups of animals including cnidarians, Platyhelminthes and echinoderms.

Q.15 Methods of Asexual Reproduction in Parthenogenesis?

Parthenogenesis:

The process of production of an organism from an unfertilized egg.

Patterns of Parthenogenesis:

There are many patterns of parthenogenesis.

- Two most common:
  (i) Meiotic parthenogenesis
  (ii) Ameiotic parthenogenesis

(i) Ameiotic Parthenogenesis:

- In this eggs are diploid
- They are produced without meiosis.
- Young ones are the clones of their parents
- This method is common in flatworm, rotifers, crustaceans and insects

(ii) Meiotic Parthenogenesis:

- Eggs are produced by the meiosis process
- Haploid egg starts development

- Diploid number restored by chromosomal duplication
- In some insects remain haploid, no chromosomal duplication
- Egg may be activated by sperm to start parthenogenesis.
- Here sperm will not fertilize egg, just activates
- In other egg activated spontaneously

Q.16 Write advantage and disadvantage of parthenogenesis. And parthenogenesis in insects and reptiles also?

Advantages of Parthenogenesis:

- When population consists of only females
- When environment is very stable
- Social organization of colonies of bees, wasps, ants (for males)

Disadvantages of Parthenogenesis:

- Less genetic variability
- Little chances for organisms to shift gene combination in changing environment

Parthenogenesis in Insects:

Very common in wasps bees and ants.

Parthenogenesis in Reptile:

- Parthenogenesis have been observed in some lizards and snakes.
- New Mexico Whiptail lizard are exclusively females.

Q.17 Write advantages and disadvantages of Asexual Reproduction ?

- **Advantages of A sexual reproduction:**

1.Species Survival:

Asexual reproduction is specially the beneficial to those species have one sex only.

2. Minimal Energy Requirements

Only one sex is involved so less amount of energy required

3. Positive Genetic Combinations Passed on to Next Generation

Because there is no genetic recombination, the good traits acquired of an organism are transmitted to next generation

4. Rapid Increase in Population

- Under stable conditions animals reproduce asexually causing a rapid increase in population

□ Disadvantages of Asexual Reproduction:

1. Low Genetic Variability:

- No genetic recombination
- Very little variations
- Population is genetically same

2. Susceptibility Against Diseases :

- As population is genetically same, so equally susceptible to disease or changing environmental conditions
- This may result in the death of whole population at once.

3. All Mutations Pass on to Next Generation:

- Most of the mutations are detrimental
- In sexual reproduction only germ cell mutation are passed on
- In Asexual reproduction all mutations are passed on to next generation

Q.18 Write Sexual Reproduction?

Define:

It the types of reproduction in which new offspring are produced by the fusion of gametes.

Two Sexes are Involved:

- Normally two sexes are involved:
- Male and female
- One set of chromosomes from male other from female
- New combinations formed

Types of Gametes:

Two types of gametes

(i) Male gametes: sperms
(ii) Female gametes: eggs

- Both are haploid and produced by the process of meiosis

Production of Gametes:

- Gametes are produced in gonads
- Male gonads: testes , Female gonads: ovaries
- Sperms produced in testes: Spermatogenesis
- Eggs produced in ovaries: Oogenesis

Q.19 differentiate between external and internal fertilization?

External Fertilization	Internal Fertilization
☐ In which fusion of gametes occurs outside the body of animals. • Normally in water • Broadcast spawning • Fertilization is external, development is also external	☐ In which fusion of gametes occurs inside the body of animals. • In mating process male introduces sperms into female body, sperms are motile they approach to female gametes

Spermatophores

- Many sperms are packed into a ball like structure called spermatophore
- These spermatophores are introduced into female reproductive tract
- Sperms are protected in spermatophore
- Minimal loss of sperms

Number of Gametes Produced

- Species with external fertilization produced large number of gametes
- Species with internal fertilization produce relatively few gametes

Q.20 Explain Hermaphroditism..?

It is the condition in which both male and female sex organs are present in the same animal.

Hermaphrodites

Beneficial to sessile animals

Monoecious:

- Gr. *Monos,* single and *oikos,* house
- One room for two sexes
- Means both sex organs are present in one animal.
- Hermaphrodites are also called monoecious

Fertilization in Hermaphrodites:

- Some undergo self fertilization
- But mostly cross fertilization
- Two animals come close to each other, both donate and receive sperms

Sequential Hermaphrodites:

- When animal is one sex in first phase of life and opposite sex in other phase of life.

- Means animal is male in first phase of life and later becomes female, vice versa

Protogyneous	Protandrous
• Those animal which are female in first half of life. • In 2nd phase of life they are male	• Those animals which are male in first half of life. • In 2nd phase of life they are female

Q.21 Explain examples of Hermaphroditism in details?

1. Earthworm
2. Mangrove killifish
3. Hamlets
4. Sea Slugs
5. Clownfish
6. Wrasses

1. Earthworm:

Earthworms and all oligochaetes are monoecious .2 pairs of testes one pair of ovaries

During copulation two worms line up and facing opposite direction and exchange gametes

2. Mangrove Killifish:

It lives along the costs of North, South and Central America.They routinely reproduce by self fertilization

3. Hamlet:

It is a fish mainly found in coral reefs of Crabbean sea and Gulf of Mexico,They do not practice self fertilization

4. Clone Fish:

Colorful reef fish found in symbiosis with sea anemone.They are protandrous.The large female and one reproductive male.When female dies reproductive male turns to female.

5. Wrasses:

Group of coral fishes in which protogyny is common.

Q.22 Write Bisexual Reproduction..?

It is the type of reproduction in which two separate individuals are involved Male and female

Dioecious:

Gr. *Di,* two and *oikos*, room.Two separate rooms for two sexes

Bisexual Reproduction is More Common :

.Most of the animals have bisexual reproduction method

One benefit of this is that the offspring have new gene combination and new genotypes appear

Sexual Dimorphism:

It is the condition in which both male and female are phenotypically different.The difference is mainly in

- Secondary sex characters
- Size, colour, weight

Q.23 Write Advantages and Disadvantages of Sexual Reproduction?

Advantages:

a) Genetic Variability:

- Genetic recombination
- New combination of traits
- This may help in changing environment or in case of disease

b) Foundation for Evolution:

- Genetic variability is there
- Variation is the foundation of evolution

c) Elimination of Deleterious Traits:

- As there is genetic recombination
- Deleterious mutation are eliminated

Disadvantages

a) Asexual animals:

- Not feasible for animal which rely on asexual means only
- Especially population with one sex only

b) Adaptive Gene Combination Dismantle:

- In sexual reproduction new genetic combination arise
- This will upset the already established adaptive gene combination

c) Wastage of energy:

- Many of the gametes are not fertilized
- This will cause a significant loss of energy

Q.24 Write Reproductive Strategies in Amphibians in details?

1. Fertilization
2. Eggs
3. Breeding
4. Courtship behaviour
5. Development

1. Fertilization:

- **Order Caudata:**

  Mostly internal fertilization

- **Order Gymophiona:**

  Internal fertilization

- **Order Anura:**

External fertilization

2. Eggs:

- No hard covering
- Order Caudata
- Single, clumps, strings
- Order Gymophiona
- Retained within body, some lay eggs
- Order Anura
- Laid in water, moist soil, tree trunks

3. Breeding:

- **Temperate region**

Linked to temperature

- **Tropical region**

Linked to rainy season

4. Courtship Behviour:

- Helps to locate breeding sites
- Identify potential mates
- Prepare individuals for reproduction
- U Ensure egg fertilization
- Olfactory, Visual, Vocalization and tactile cues

5. Development:

- **Order Caudata**

Larva aquatic, adult terrestrial

- **Order Gymophiona**

In oviduct with fetal teeth to feed from lining, others aquatic larvae

- **Order Anura**

Aquatic larvae, herbivores

Q.25 Explain Reproductive Strategies in Reptiles and Birds in details?

1. Fertilization
2. Eggs
3. Breeding
4. Courtship behaviour
5. Development

1. Fertilization :

- Reptiles and birds are dioecious
- Fertilization is internal
- Fertilization in the upper portion of oviduct
- sperms may be stored in seminal vesicles up to 4years in some turtle and 6years in some snakes

2. Eggs:

- Soft to Hard shells
- Desiccation resistant
- Enough amount of yolk
- Amniotic eggs with three extraembryonic membranes
    - Amnion
    - Chorion
    - Allantois

3. Breeding:

Reptiles:

- Copulation in winter during hibernacula
- Fertilization in spring

Birds: In spring

4. Courtship Behviour:

Reptiles:

- Complex courtship behaviour
- Male actively seek out female
- Courtship function for sex recognition
- head-bobbing males lizards
- Tactile in snakes

Birds:

- Territorial behaviour
- Attract females by sounds (wood pecker), dancing (cranes, peacocks)
- >90% monogamous
- Polygamy in species where young ones require less parental care

5. Development

Reptiles:

- External,
- Fertilized eggs remain unattended
- In some cases eggs are cared for humidity
- In crocodile post hatching care of dependent young ones.

Birds:

- External,
- Eggs are incubated, protected from predators
- Extensive parental care of young ones

Q.26 Explain Reproductive Strategies in Mammals?

1. Eggs:

- Amniotic eggs
- Little amount of yolk except egg laying mammals

2. Fertilization :

- Mammals are dioecious
- Internal fertilization
- Some bats have delayed fertilization
- Fertilization in the upper portion of oviduct

3. Courtship Behviour:

- Complex courtship behaviour
- Male actively seek out female
- Territorial behaviour
- Vocalization is very common

4. Reproductive Cycle:

- Most mammals have definite time for egg maturation
- Females undergo estrus cycle: female is behaviorally and reproductively receptive to males. Monoestrus and polyestrus

5. Development:

Mostly viviparous

Montremes: oviparous

Marsupial mammals: uterine milk, primitive placenta, short gestation period

Eutherian mammals: embryo implanted deeply, placenta formation, gestation period 20 days to 19 months

Q.27 Examples of Reproductive Function in Fishes and Amphibians?

1. Bony fishes
2. Amphibians

1. Bony fishes:

1000's of eggs are fertilized. Fungal and bacterial infection

- Siltation
- Predation
- Yolk only for short time development.
- Hatched young ones actively feed.
- Achieve maturity in short time.
- In some fishes embryos are cared properly.
- Sea horse and pipefish males carry embryo in their ventral pouches

2. Amphibians:

- Mostly external fertilization in water in few internal
- Eggs are cared aeration, and infected eggs are removed from clutches
- *Pipa* female carry eggs on back
- Little amount of yolk
- Development in much longer time than amphibians fishes
- Larval stage to complete development
- Metamorphosis
- *Rheobatrachus* females carry egg and tadpole in stomach

Q.28 Examples of Reproductive Function in Reptiles and Birds?

1. Reptiles:

- True terrestrial vertebrates
- Shelled eggs with extraembryonic membranes
- Chorion amnion… protect and provide aqueous environment
- Allantois… gaseous exchange, excretory products
- Large amount of yolk

- Complete development within shelled egg
- No larval stage
- Some snakes are ovovivparous

3. Birds:

- Except water fowl lack penis
- Deposit semen against cloaca
- Egg shells thicker than reptiles
- Fertilization before shell formation
- Hard shelled eggs are incubated by parents
- Extensive parental care
- Parental care and feeding more common than amphibians, reptiles and mammals

Q.29 Examples of Reproductive Function in Mammals?

1. Monotremes
2. Marsupial Mammals
3. Placental Mammals

1. Monotremes:

- Duckbilled platypus, spiny anteater
- Lay eggs
- Duckbilled platypus…lay eggs in its nest/burrows
- Spiny anteater… incubate eggs in ventral pouch

2. Marsupial Mammals:

- Koala, kangaroo
- Gestation period: 8 to 40 days
- Suckling….. 60 to 120 days

3. Placental Mammals:

- Most of the mammals
- Deer, lion, monkeys etc.
- Internal development
- Placenta formation

- Young ones nourished through mammary glands

Q.30 Write Reproductive Patterns in Animals?

1. **Reproductive Patterns:**

- The pattern of egg laying and development of young ones.
- In all invertebrates and vertebrates one of the three patterns.
- Oviparity
- Ovoviviparity
- Viviparity

2. Oviparous:

- Which lay eggs and development outside the body of animals
- Fertilization may be internal or external
- Majority of invertebrates and many vertebrates are oviparous
- Parental care may be or may not

3. Ovoviparous :

- Which eggs are retained into the body but the young one gets all its food from yolk of egg
- Fertilization is internal
- Development also internal
- **Invertebrates:** Annelids, brachiopods, gastropods
- **Vertebrates:** Fishes and reptiles

4. Viviparous:

- Which eggs develops into the body and the young one gets its nourishment from mother
- Fertilization is internal
- Development also internal
- **Invertebrates:** Scorpions

- **Vertebrates:** most mammals and elasmobranch fishes, few amphibians and reptiles

Q.31 Write Invertebrate Reproductive system with male and female system?

1. Invertebrates with no reproductive organs:

- Some invertebrates like polychaetes have no permanent reproductive organs
- Body cavity lining cells produce gametes
- Which are released coelomic or nephridial duct, some times body wall may rupture

2. Male reproductive system:

Parts of male reproductive system

- Testes
- Sperm duct/vas deferens
- Seminal vesicle
- Ejaculatory duct
- Accessory glands

3. Female reproductive system:

Parts of female reproductive system

- Ovaries
- Oviducts
- Seminal recepactacle
- Bursa
- Gonopore
- Oviposter

Q.32 Write Human Male Reproductive System?

1. Functions of Male Reproductive System:

- Two basic function:
- Production of sperms
- Transfer of sperm to female reproductive tract

2. Components of Male reproductive system:

- Scrotum
- Testes
- Rete testis
- Epididymis
- Duct Deferens
- Ampulla
- Seminal Vesicle
- Ejaculatory Duct
- Prostate Glands
- Bulbourethral Gland
- Penis

Q.33 Explain Production of Sperms in human?

1. Testes

- A pair of testes
- Scrotum
- 34 ^{o}C for sperm production and survival

2. Seminiferous Tubules:

- About 800 in each testes
- **Seminiferous tubules walls contain Two types of cells:**
    - Spermatogenic cells
    - Sustentacular cells
- Clusters of interstitial cells between seminiferous tubules.

3. Spermatogenesis:

- Thousands of sperms are produced in each second in seminiferous tubules
- The process of sperm production is called spermatogenesis
- Spermatogenic cells are present in the wall of seminiferous tubules
- Spermatogenic cells are also called spermatogonia
- Spermatogomium increase in size and becomes primary spermatocyte
- Meiosis-I results in two secondary spermatocytes
- After meiosis-II four haploid spermatids are formed
- Spermatid undergo morphological changes and are transformed into sperms which are motile

4. Structure of Sperm:

A mature sperm consists of:

- **Head:** contain haploid nucleus and acrosome cap having acrosin.
- **Midpiece:** have mitochondria
- **Tail:** microtubules that bend for whiplike movements

Q.34 Explain transport of Sperms in human?

1. Seminiferous Tubules:

- Seminiferous tubules contain spermatogenic and sustentacular cells.
- Spermatogenic cells produce sperms while sustentacular cells secrete fluid for nourishment of sperms.

2. Rete Testis and Epididymis:

- Sperms from seminiferous tubules comes into a network of called rete testis
- Then into epididymis where:
- Sperms are store
- Muscular contraction propel sperm forward

3. Duct Deferens and Ampulla:

- Sperms through duct deferens are transported to ampulla where stored till ejaculation.

4. Seminal Vesicles:

- Secrete water, fructose, prostaglandins and Vitamin C
- This is energy source for sperm motility and helps to neutralize pH of female reproductive tract

5. Prostate Gland:

- **Secretes water, enzymes, cholesterol, buffering salts and phospholipids**

6. Bulbourethral Glands:

- Secrete clear alkaline fluid that:
- Lubricates urethra to facilitate ejaculation

7. Semen:

- Fluid that results from the combination of sperms and glandular secretions.
- Average human ejaculation contain 3 to 4 ml of semen having 300-400 million sperms**.**

Q.35 Describe Hormonal Control of Male Reproductive Function?

1. Androgens and Gonadotropins:

- Male sex hormones are collectively called androgens e.g., Testosterones
- The hormones which travel from brain and pituitary to testis and ovaries are called gonadotropins e.g., FSH and LH

2. Testosterone and Inhibin:

- Testosterone: secreted by interstitial cells
- Role in development of testis, prostate and secondary sex characteristics
- Inhibin: Produced by sustentacular cells
- It inhibits FSH

3. Hormonal Control:

- Low level of testosterone is detected by hypothalamus
- Hypothalamus release GnRH
- GnRH effect the pituitary gland to release FSH and LH (ICSH)
- FSH stimulate spermatogenesis
- LH stimulate the production of testosterone
- Testosterone inhibits the secretion of LH
- Inhibin inhibits the secretion of FSH

Q.36 Explain Human Female Reproductive System?

1. Functions of Female Reproductive System:

- Gamete production
- Nourish, carry and protect developing embryo
- Nursing young one after birth

2. Structures of Female Reproductive System:

- **Ovaries:** produce eggs, estrogen and progesterone
- **Uterine tubes:** carry eggs ovaries to uterus. Fertilization here.
- **Fimbriae:**
- **Uterus:** embryo implanted here
- **Vagina:** receives semen from male and baby passes through it during birth
- **External genitalia:** sexual arousal and protection
- **Mammary glands:** produce milk

Q.37 Explain menstrual Cycle in Human?

1. The Menstrual Cycle:

- It is the monthly cycle of changes in ovaries and preparation of uterus for implantation of embryo
- The cycle begins at puberty and stops at the age of 45 to 55 years

2. The Follicular Phase:

- The controlling center is hypothalamus
- It releases GnRH which stimulate the release of FSH and LH from anterior pituitary
- FSH starts the development of oocyte in the ovarian follicles
- Developing follicle produces estrogen
- Estrogen builds up and prepares uterus lining to become thicker (form endometrium) for implantation of embryo

3. Ovulation:

- Elevated level of estrogen increases the release of LH from pituitary gland and inhibits FSH
- LH cause the mature follicle to enlarge rapidly and release secondary oocyte
- This process is ovulation

4. The Luteal Phase:

- Mature follicle after release of oocyte becomes corpus luteum
- Corpus luteum secretes estrogen and progesterone.
- They complete the development of endometrium and maintain it for 10-14 days

5. Menstruation:

- If the oocyte is not fertilized, corpus luteum disintegrates
- Estrogen and progesterone production stops
- As a result endometrium breaks and menstruation starts

Q.38 Write Follicle Development in Human?

1. Structure of Ovaries

- In cross section every ovary consists of:
- **Medulla:** have blood vessels and nerves

- **Cortex:** have developing follicles

2. Primordial Follicles:

- During first five months of development
- Many small follicles are formed in the ovary of female fetus
- These are called primordial follicles

3. Primary Follicles:

- At the time of puberty during each menstrual cycle some primordial follicles began to develope further and become primary follicles
- Then secondary follicles and graafian follicle

4. Ovulation:

- Many follicle start development in each cycle but only one completes
- The follicle before release of egg is called graafian follicle
- LH hormone causes the rupture of follicle to release egg

5. Corpus Luteum and Corpus Albicans:

- Graafian follicle after release of oocyte becomes corpus luteum
- Corpus luteum remain active for 10-14 days
- If fertilization does not occur corpus luteum forms corpus albicans

6. Atretic Follicles:

- During each menstrual cycle several follicles start development but only one matures
- Others disintegrates by the process atresia
- During this cells are disintegrated by apoptosis.

Q.39 Explain Production and Transport of Egg in Human?

1. Egg Production:

- Egg production starts in female while still at the developmental stages.

- A definite number of eggs development takes place in specific structures called follicles
- Egg development is halted at early stage of meiosis-I
- At puberty during menstrual cycle meiosis-I is completed and meiosis-II is started
- Meiosis-II will be completed after fertilization

2. Ovulation:

- It is the release of oocyte from ovary.
- LH cause the release of oocyte from ovary

Egg Entry into Fallopian Tubes:

- Ovary is encircled by finger like ciliated fimbriae
- The ciliary action and muscular contraction cause the entry of oocyte into fallopian tube also called uterine tube

3. Fertilization:

- Fertilization takes place in upper third portion of uterine tube.
- Oocyte completes meiosis-II after fertilization
- Fertilization should take place within 24 hour of ovulation

4. Movement of Egg in Uterine Tube:

- Egg moves due to ciliated cells and muscular contraction of uterine wall
- Takes about 5 days for uterine tube to uterus
- If fertilized, embryo implanted into uterus
- If unfertilized, released out

Q.40 Write hormonal Control of Female Reproductive Function?

1. Hypothalamus and Pituitary Gland:

- Low level of estrogen stimulate hypothalamus to release Gonadotropin–releasing hormone (GnRH)
- GnRH acts on pituitary gland to release follicle stimulating hormone (FSH) and luteinizing hormone (LH)

2.Follicle Stimulating Hormone:

FSH and LH from pituitary act on ovary. FSH start the development of follicle. 15-20 follicles start development but only one matures..Ovary also starts to produce estrogen

3. Estrogen:

Estrogen causes the thickness of endometrium of uterus.Higher level of estrogen triggers LH and inhibit FSH form pituitary gland.

4. Luteinizing Hormone:

LH from pituitary causes the ripening of ovarian follicle and release of oocyte from follicle.LH also convert collapsed follicle into corpus luteum

5. Progesterone:

Corpus luteum secretes estrogen and progesterone. Both complete development and maintain endometrium. Both also inhibit LH and FSH. If no fertilization, FSH and estrogen level drop new cycle starts.

Q.41 Explain historical Perspective of Embryology?

1. Preformation Theory:

Most biologist of 17th and 18th century believe in it. This theory says that gametes contain miniature version of all the element present in an adult

2. Epigenesis Theory:

This theory gained popularity in mid 18th century.This theory says that embryo contain material from which embryo is gradually build.Much like a carpenter builds a building from nails, lumber and bricks

3. Birth of Modern Embryology:

Modern embryology emerged with the proposed germ layer theory.Embryo form 2-3 layers from which different body parts are formed

4.Development Biology:

It the study by which animals and plants grow. Developmental biology uses the tools of molecular biology to investigate the genetic control of development.

Q.42 Explain Gametogenesis: Spermatogenesis?

1. Spermatogenesis:

The process of sperms production in male gonads through meiosis is called spermatogenesis

2. Location and duration:

Spermatogenesis takes place in testes.In human the entire process of spermatogenesis and its transportation takes about 3 months.However, it is different in different animals

3. Spermatogoium and Primary Spermatocytes:

Testes contain diploid stem cells.Sperm mother cell is called spermatogomium.It enlarges in its size and become primary spermatocyte

4. Secondary Spermatocytes and Spermatids:

Primary spermatocyte undergo meiotic cell division.After meiosis-I two secondary spermatocytes are formed.Meiosis-II results in four haploid spermatids

5.From spermatids to sperms:

Spermatids do not undergo any further cell division.They undergo morphological changes to produce sperms.

Q.43 Write Gametogenesis: Oogenesis?

1. Oogenesis:
The process of egg production in female gonads through meiosis is called oogenesis

2. Location of Oogenesis:

Oogenesis takes place in ovaries.Follicles are specific structures in which egg formation and maturation takes place

3. Oogoium and Primary Oocytes:

Ovaries contain oogonia.Oogonium increase its size ….primary oocyte .Primary oocyte is diploid cell with duplicated chromosome.Each primary oocyte will form 1 ovum and polar bodies after reductive division

4. Secondary Oocytes and Ovum:

Primary oocyte undergo meiotic cell division.After meiosis-I one secondary oocyte and one polar body is formed.Meiosis-II results in a haploid ootid and another polar body .Ootid undergo differentiation process to form ovum.

Q.44 Write types of Eggs in details?

Classification of Eggs:

Eggs are classified on the basis of:

- Amount of yolk

Alecithal
Microlecithal
Mesolecithal

Macrolecithal

Eggs are classified on the basis of:

- Distribution of yolk

Homolecithal

Centrolecithal

Telolecithal

1.Alecithal Eggs:

They contain negligible amount of yolk **e.g.,** mammals

2.Microlecithal Eggs

They contain very little amount of yolk. **e.g.,** sea urchin, amphioxus and marsupial mammals

3.Mesolecithal Eggs:

These are the eggs with moderate amount of yolk **e.g.,** lung fish, amphibians

4.Macrolecithal Eggs:

Eggs with large amount of yolk **e.g.,** reptiles and birds

5.Homolecithal eggs:

Yolk is equally distributed **e.g.,** sea urchin, mammals

6.Centrolecithal Eggs:

These are the eggs with yolk in the center while cytoplasm in periphery **e.g.,** insects

7.Telolecithal Eggs:

Yolk is present at one end of the egg while cytoplasm at other end.Animal pole and vegetal pole **e.g.,** frogs, reptiles, birds.

Q.45 Expalin fertilization process?

Defining Fertilization:

Fertilization is the process in which egg and sperms fuse to form a diploid zygote.Zygote is the first representative of next generation

Methods of Fertilization:

External fertilization:

Internal Fertilization:

- Copulation
- Cloacal connection
- Via spermatophore

Vitelline Membrane:

It is the structure surrounding plasma membrane of egg.It is made up of fibrous proteins and receptors needed for sperm binding

Fusion of egg and sperm:

The sperm contain an acrosomal head.Acrosome have specific enzymes called lysins which dissolve egg membrane during ferlization..Many sperms approach to egg but only one successful.In some species only nucleus fuses.In others, mitochondria, centriole and flagella of sperm may get entry into egg.

Egg Activation:

A series of biochemical changes in egg which ensure the completion of fertilization and initiate embryonic development.

Q.46 Fertilization: Metabolic and Nuclear Events?

1. Raise in Intracellular pH:

After fertilization ionic changes occur in cytoplasm.This causes an increase in intracellular pH.This results a change in zygote physiology to start development

2. DNA Replication:

zygote starts the cleavage (cell division) process..For this, DNA replication starts as two copies of DNA are required for two daughter cells

3. Protein synthesis:

Rate of protein synthesis is also boosted .This is because more and more proteins and enzymes are required for cell division and daughter cells.mRNA already present in the cell is activated to start protein synthesis

4.Maternal Dominance:

The mRNA in zygote used to protein synthesis is maternal in origin.Thus bulk of initial proteins is synthesized by maternal mRNA.This influence is called **maternal dominance**

Q.47 Define Cleavage and write Its Patterns?

Defining Cleavage:

- Cleavage refers to cell divisions that occur during embryonic development.

Animal Pole and Vegetal Pole of Egg:

Animal pole: The region of egg containing cytoplasm, its organelles and less amount of yolk

Vegetal Pole: the region of egg containing more amount of yolk and less cytoplasm

Synchronous and Asynchronous Division:

Initially all the cells of embryo divide at the same time this is called **synchrony**.

As development proceeds, all cells don't divide simultaneously. This is called **asynchrony**

Distribution of Yolk Influence Cleavage Pattern:

Eggs with evenly distributed yolk have cleavage pattern that result uniform size blastomeres

Egg with unevenly distributed yolk result in unequal size blastomeres

1. Holoblastic Cleavage:

Cleavage type in which egg is completely divided into two cells.Egg contain little or moderate amount of yolk.Eggs of Amphibians, mammals, echinoderms, flatworms, nematodes

2.Meroblastic Cleavage:

Cleavage type in which egg is not completely divided into two cells.This is due to large amount of yolk.These embryo develop around or on top of yolk.Eggs of birds and reptiles

3. Spiral Cleavage:

It is pattern in which cells of animal pole are not exactly laying above the vegetal pole cells

Cleavage takes place at oblique angle forming a spiral pattern.Example: protostomes

4. Radial Cleavage:

It is pattern in which cells of animal pole are exactly laying above the vegetal pole cells. Upper tier of cells is exactly on the lower tier of cells.Example: deurostomes

Q.48 Explain Echinoderm Embryology?

General Concepts

1. Extensively Studied:

Echinoderms include starfish, sea cucumber, brittle star and sea urchin.Echinoderms are extensively studied group .Easily maintained in laboratory

2. Echinoderm Egg:

Eggs are laid in water.Very little amount of yolk.Microlecithal egg.External fertilization

3.Developmental Stages:

*Fertilization in water *Morula * Gastrula

*Pluteus larva *Development completed in approx. 30 hours

Plan of Division:

First two cleavage are from animal to vegetal pole (meridional) and perpendicular to each other

Third cleavage is equatorial and right angle to first two. Separate animal pole and vegetal pole

Fourth cleavage is medional

Q.49 Explain Morulation and Blastulation in Echinoderm Embryology?

Morulation and Blastulation

Morula:

- Cleavage results in a ball of cells called morula
- Morula is a solid ball of cells without any cavity

Blastocoel Formation:

After murula formation when embryo is about 120 cell stage, a fluid filled cavity **blastocoel** is formed in the embryo. Cell form a single layer around the blastocoel.Embryo at this stage is called blastula.

Cilia Development:

Development up to blastula stage takes place within fertilization membrane.Cilia develop at blastula stage.It comes out of fertilization membrane and start swiming

Primary Mesenchyme:

At late blastula stage, some cells breakoff and position themselves within blastocoel..These cells are called primary mesenchyme..They will form skeletal elements (spicules) of embryo.

Q.50 Explain Gastrulation in Echinoderm Embryology?

Gastrulation

Defining Gastrulation:

Gastrulation is the process in which blastula is changed into gastrula. During this process primary germ layers are formed

Cells Invagination:

Late blastula is about 1000 cells of different size.Gastrulation begins at a point in vegetal half

Where cells start invagination

Blastopore and Archenteron:

The point where invagination starts is blastopore.The invagination forms a new cavity in the embryo called **archenteron**.Archenteron gradually replaces blatocoel

Blastopore and Archenteron Formation:

Blastopore will form anus .Archenteron is future gut of echinoderms

Secondary Mesenchyme Cells:

Some cells at the tip of developing archenteron get apart and form secondary mesenchyme cells

Secondary mesenchyme cells play a central role in pulling archenteron up to blastocoel wall

Front end is called anterior end

Pluteus Larva :

After gastrulation the embryo attain prism shape then pluteus larva.Pluteus larva is plaktonic

It has bilateral symmetry.Although adults have radial symmetry

Q.51 Explain Amphibian Embryology?

General Concepts

Amphibian Eggs:

Mostly laid in water . External fertilization. Mesolecithal egg. Animal and vegetal poles

Pigmented animal pole. Heavy yolk in vegetal pole

Protection of Amphibian Egg:

Usually little care by parents.Darkly pigmented animal pole on upper side camouflage from predators.Light colour vegetal pole blends with sky when viewed form below

Developmental Stages:

Fertilization in water Morula Blastula

Gastrula Larval stage Adult

Q.52 Explain cleavage in Amphibian Embryology?

Cleavage

1.Holoblastic Cleavage:

Amphibians have mesolecithal eggs.Cleavage completely divide cell in two.Due to yolk at vegetal pole cleavage is slow.Size of cells at animal pole is smaller while larger at vegetal pole

2. Radial Cleavage:

Cleavage lines pass in such a way that cells lie exactly one above the other.Radial cleavage is the characteristic of deuterostomes

3. Plan of Division;

First cleavage start from animal to vegetal pole.Cleavage at vegetal pole is slow due to large amount of yolk.It also divides gray crescent from middle.Second cleavage is perpendicular to the first and divides embryo into four.Third cleavage is equatorial and right angle to first two. Separate animal pole and vegetal pole.Fourth cleavage is medional

Q.53 Explain Morulation and Blastulation and gastrulation in Amphibian Embryology?

Morulation and Blastulation , Gastrulation

1.Morula:

Cleavage results a ball of cells called morula.Morula is a solid ball of cells without any cavity.Cell towards animal pole are smaller while larger at vegetal pole due to presence of yolk

2.Blastocoel Formation

After morula formation, a fluid filled cavity **blastocoel** is formed in the embryo.Cell form multiple layers around the blastocoel.In amphibians blastocoel is formed towards animal pole due to yolk at vegetal pole

Gastrulation in mphibian Embryology

Defining Gastrulation:

Gastrulation is the process during which cells arrange themselves into different layers called primary germ layers

Blastopore Formation:

Gastrulation begins with a slit like opening **blastopore** between gray crescent and ventral pole.

Animal pole margin of blastopore is called its dorsal lip.Cells at the bottom of groove move anteriorly and groove spreads transversely

Yolk Plug Formation:

The ring like blastopore surrounds the yolk cells and they protrude out of the blastopore .This is called yolk plug.Later blastopore cells enclose yolk plug and blastopore is considered as closed.

Epiboly:

These ectodermal cells towards blastopore and spread over entire embryo. This is called **epiboly**

Mesoderm Formation :

Lateral to notochord mesodermal cell spread and thicken along sides of embryo.These thickenings are called **somites**.Somites appear as row of pumps from side view .Eventually somites spilt to form coelom

Q.54 Explain Neurulation in amphibian Embryology?

Neurulation:

The process of forming neural tube is called neurulation

Neural Plate:

An oval shape area on the dorsal surface of embryo is formed.Microfilament flatten and thicken this area.This region is neural plate

Neural Folds:

The edges of the neural plate roll up and over the midline of neural plate.These raised edges in longitudinal direction are called neural folds

Neural tube;

The raised neural folds .Meet with each other longitudinally over the midline of neural plate

Now neural plate is converted into a tubular structure called neural tube

Brain :

The portion of neural tube that is close at the end will form brain

Q.55 Explain Avian Embryology and structure of avian egg ?

General Concepts

1. Chick Embryo:

Chicken development is used as a model for the development of birds and reptiles

2. Egg Types:

Chicken egg contain large amount yolk .They are macrolecithal eggs.They are also telolecithal eggs.Yolk is also protected by albumin $CaCO_3$ shell

3. Fertilization:

Fertilization is internal.In upper portion of oviduct

4. Developmental Stages:

Development starts right after fertilization while egg is still in oviduct.All developmental stage are completed within shelled egg.In chicken development is completed in approximately 3 weeks

Structure of Avian Egg

1.Egg Formation:

In ovaries Large amount of yolk. Mature egg released from ovaries .Comes into oviduct.Fertilization takes place in upper portion of oviduct.

2. Egg White:

Membranes and fluid collected around the egg while moving through oviduct. Water and albumin is deposited around the egg. This protect from desiccation and mechanical damage. Albumin is used for protection and source of nutrition. Two denser strands of albumin attached to the egg shell

3. Egg Shell:

Shell gland secrete $CaCO_3$ which deposit around the egg. It is also impregnated with proteins. Thousands of tiny pores (40-50µm)are for exchange of gases .Inside shell two membranes. Air pocket.

Q.56 Write cleavage , Blastulation and gastrulation in avain embryology?

Avian Embryology: Cleavage

1.Meroblastic Cleavage:

Chicken eggs have large amount of yolk. Cleavage do not divide cell in two. There is no division in the large amount of yolk. Discoidal cleavage

2. Radial Cleavage:

Cleavage lines pass in such a way that cells lie exactly one above the other.Radial cleave is the characteristic of deuterostomes

3. Plan of Division:

Same like other deuterostomes. First two cleavage are vertical and perpendicular to each other. Third cleavage is equatorial and right angle to first two. Fourth cleavage is vertical

Avian Embryology: Blastulation

1. Blastoderm:

- The cells in the animal pole dived
- Approximately 60,000 cell form blastoderm

2. Blastocoel:

- The yolk in below blastoderm is used and a fluid filled cavity appears
- This cavity is called blastocoel

3. Epiblast and Hypoblast:

- Blastoderm splits into two layers
- Outer layer is called epiblast
- Inner layer is hypoblast

Avian Embryology: Gastrulation

1. Primitive Streak:

- The dorsal cells of blastoderms form a medial linear invagination called primitive streak
- Primitive streak gradually extends anteriorly
- It defines the axis of embryo
- It also separate right and left side

2. Henson's Node:

The anterior margin of primitive streak there is a regional thickening called Henson's node.Henson's node contains funnel shaped depression through which cell enter between epiblast and hypoblast.

The ectodermal cell migrate and enclose the yolk .This process is called **epiboly**

Q.57 Explain Mammalian Embryology?

General Concepts

1.Mammalian Eggs:

- Egg laying mammals… large eggs
- Marsupial mammals… microlecithal eggs
- Placental mammals… alecithal eggs
- Placental mammals egg is smallest in animals kingdom

2. Fertilization and Development:

- Internal fertilization
- Placental and marsupial mammals have internal development
- Both give birth to young ones
- Marsupial mammals have premature birth
- Keep in marsupium
- Placental have long gestation period
- Fetus is fed through placenta

3. Developmental Stages:

Zygote Embryo Fetus newborn

Q.58 Write Cleavage and Blastulation and gastrulation in Mammalian Embryology?

Cleavage:

Mammalian cleavage is slowest in animal kingdom.Blastomeres have unique orientation.Rotational cleavage. First cleavage is meridional while in second cleavage one is meridional and other horizontal. Cell are not 2-4-8, but may be odd number

1.Compaction:

Through 8 cell stage, loose arrangement with plenty of space. After third cleavage the cells tightly huddle together to form a compact ball.

2.Morula:

8 cells divide to produce 16 cell morula

A small group of internal cells is called inner cell mass (ICM) while the outer cells form trophoblast.ICM form embryo while trophoblast forms chorion for placenta formation

3.Totipotent and Pluripotent:

8 cells stage blastomere can form both ICM and trophoblast, they are called totipotent

ICM cells can form any type of embryo cells, they are called pluripotent

Blastula:

Trophoblast cells of morula secrete fluid that create blastocoel. As blastocoel expands ICM and trophoblast take their positions. At this stage blastula is called blastocyts

Gastrulation:

1.Similarity with Reptiles and Birds:

Both mammals and birds originate from reptile. Despite difference in amount of yolk both have striking similarity in development

2. Epiblast and Hypoblast:

Hypoblast cells form extraembryonic endoderm which form yolk sac. Epiblast form embryonic endoderm and amnion cavity.

Q.59 Write hormonal Regulation in Pregnant Female?

Human Chorionic Gonadotropin (hCG):

The cells developing embryo and placenta release hCG. hCG keeps the corpus luteum from disintegration. High level of hCG is used as immunological for pregnancy by 2 weeks

Progesterone:

Progesterone is released by corpus luteum. Progesterone maintains the endometrium lining. This retains pregnancy

Prolactin:

It is secreted by pituitary gland. It is produced in pregnant female to start the process of milk production and eject milk after childbirth

Oxytocin:

It is also produced by pituitary gland. It caused contraction of smooth muscles to start labour and expulsion of young one .It also stimulate nipples to release milk

Prostaglandins:

Prostaglandins secretion increase during labour. It also causes smooth muscle contraction

Q. 60 Explain Prenatal Development in Human?

1st Trimester

Embryo, Fetus and Newborn:

After two weeks of fertilization till the end of 8th week is embryo. From 9th week till birth is fetus. After the birth is newborn or baby

Timesters:

Total gestation period is 280 days .This duration is divided into 3 months periods called trimesters

First Trimester:

Start from fertilization

Zygote divides to become morula

By 4th day it is converted into 50- 120 cell blastocyst

Implantation:

As embryo reaches into uterus, it implants itself onto uterine wall. Trophoblast invade the endometrium

Gastrulation:

During gastrulation embryonic layers are formed. Inner cell mass give rise to embryo. Gastrulation is similar to birds and reptiles

Organogenesis and Growth:

Once gastrulation completed organogenesis starts. Different organ systems are shaped. By middle of first trimester all body organ begin to develop. Embryo also grows in size.

Q.61 Write 2nd Trimester and 3rd Trimester in prenatal Development in Human?

Prenatal Development in Human: 2nd Trimester

Second Trimester:

It ranges from 14 to 27 week

Major Events of Second Trimester:

Rapid growth of fetus

- **14 week**… RBC formation, sex apparent
- **16 week**….eyes move
- **17 week**….Toenail develop, heart beat
- **18 week**…. hearing
- **20 week…..** Feel movement, sleep wake
- **21 week…** baby sucks finger
- **22 week….** Eyebrow and hair visible
- **23 week….** Finger prints form
- **26 week…** Lung develop

Prenatal Development in Human: 3rd Trimester

Third Trimester:

- It start from 7th month till birth
- Fetus has potential to survive if born premature

Developmental Changes

- **27 week…..** Eyes partially open
- **30 week…** Baby hair grow
- **31 week….** Rapid weight gain
- **32 week….** Breathing practice

- **34 week….** Fingernails grow
- **35 week….** Skin pink and smooth
- **37 week…** Head turn down
- **38 week…..** Toe nail grow
- **39 week…..** Chest prominent, testis continue to descend
- **40 week….**Birth

Q.62 Explain Placenta?

Placenta is s temporary organ connects developing fetus to mother via umbilical cord

Placenta Formation:

Human have placenta formed form chorion and allantois.The tiny chorionic villi sent out from blastocyst form placenta .Placenta formation starts from day 6 and is completed by 1st trimester. Embryonic blood vessels of placenta do not merge with maternal blood vessels.It is about 22cm in length

Role of Placenta:

Placenta is used for:

- Nutrient uptake
- Thermoregulation
- Waste elimination
- Gaseous exchange
- Fight against internal infection
- Placental hormone production

Umbilical cord:

Placenta is connected to fetus through umbilical cord.It contain two fetal umbilical arteries and one fetal umbilical vein.They spiral about each other in umbilical cord.Average size is 55cm with 1-2cm diameter.

Q.63 Explain Birth in Human?

1. Total Gestation:

It is the total time period which embryo or fetus spends within body of mother. In human it is 280 days or 40 weeks

2. Parturition:

The process of birth is called parturition. During parturition, mother's uterine muscles begin to contract and cervix begins to dilate

3. Relaxin Hormone:

Mother ovaries and placenta produce relaxin hormone. Relaxin causes mother pelvic bone to slightly separate, so the baby may pass from birth canal

4.ACTH Stimulation:

During birth baby pituitary gland secretes Adrenocorticotropic hormone (ACTH) .ACTH stimulates adrenal glands to secrete prostaglandins.Prostaglandins work along with oxytocin

5. Oxytocin:

Stretch in uterine wall stimulate pituitary to secrete oxytocin hormone in mother. Oxytocin and prostaglandins cause powerful muscular contractions. Muscular contraction increases over time

Q.64 Write multiple Births?

1. Uniparous

The organisms which give birth to only one offspring at a time are called uniparous.. *E.g.,* human, cows, bufallos

2. Multiparous:

These are animals giving birth to more than one young ones. *E.g.,* mice, rat, dogs, cats

Usually the young ones are both male and female. However armadillo give birth to four young ones at same time, all males or females

3.Monozygotic Twins:

Produced from one zygote (identical twins).
They are strikingly alike and of same sex

About 33% have separate placenta.

Majority have separate amniotic sac

Small proportion have one placenta and one amniotic sac.

Indicate separation occur after 9 day of pregnancy.

Conjoining risk, condition is Siamese twining

4. Dizygotic Twins:

Produced from two zygote (Fraternal twins).

Do not have striking similarities.

May be of same or different sex.

They have separate placenta and separate amniotic sac

Q.65 Explain Milk Production and Lactation

1. Mammary Glands:

Mammary glands are unique feature of mammals. Mammary glands are evolved from sweat glands in the skin

2. Lactation:

It is the process of milk production and release by the mammary glands

3. Hormonal Control of Milk Production:

During pregnancy breasts enlarges due to hormone prolactin. Estrogen and progesterone inhibit the secretion of milk before birth. After birth both decrease and milk production starts

3. Hormonal Control of Milk Production:

Newborn stimulate pituitary to release oxytocin and prolactin. Prolactin produce milk while oxytocin triggers its release from mammary glands

4. Colostrum:

It is a higher protein fluid in breast at the time of birth. It contains maternal antibodies to boost immune system. It works as laxative to remove meconium. Mother's breast don't release milk for 1-3 days.

Q.66 Explain Extra embryonic Membranes?

1. Amniotes and Anamniotes:

Reptiles, birds and mammals embryo have amnion, chorion and allantois are called amniotes. Eggs of fishes and amphibians do not have amnion, chorion and allantois are called anamniotes

2. Role of Extraembryonic Membranes:

Amnion: encloses embryo in a fluid filled sac. Protects against shock and desiccation

Chorion: near to shell, mechanical support, highly vascular, gaseous exchange

Allantois: storage and excretion of toxic wastes, also for gaseous exchange

Yolk sac: distributes nutrients to developing embryo

3. Development of Extraembryonic Membrane:

Ectoderm and mesoderm on both side of embryo lift off the yolk and grow dorsally, they fuse to form chorion (outer side) and amnion (inner side) .

Q.67 Write Primary Germ Layers in Ectoderm derivatives ?

Ectoderm Derivatives

Primary Germ Layers:

During gastrulation three germ layers are formed

- Ectoderm
- Mesoderm
- Endoderm

These germ layers give rise to different body parts

Formation of Ectoderm:

It is the outer most layer of cells formed during gastrulation. Ectoderm is most exterior layer of embryo

Neural plate is converted into a depression called **neural groove**

Development of Epidermis:

The basal layer or germinal epithelium give rise to all epidermal cells

- Epidermis also give rise to:
  - hair,
  - sebaceous glands,
  - sweat glands,
  - Nails, claws, hoofs

Q.68 Write Primary Germ Layers in Endoderm Derivatives?

Endoderm Derivatives

1.Endoderm:

Endoderm is the inner most primary germ layer formed during gastrulation

2. Digestive Tract:

The entire gastrointestinal tract except mouth, pharynx and anus develop from endoderm.The lining of all glands which are part digestive system.Liver, gall bladder and pancreas

3. Respiratory Tract:

Lungs , Trachea , Bronchi , Alveoli

4. Endocrine Glands:

Thyroid , Parathyroid , Thymus , Pancreas

5. Auditory System:

Auditory tube , Tympanic cavity

Q.69 Write Primary Germ Layers in Mesoderm Derivatives?

Mesoderm Derivatives

1. Mesoderm:
Mesoderm is one of the primary germ layer formed during gastrulation. It is present between ectoderm and endoderm

2. Coelom Formation:

It is the body cavity surrounded by mesodermal cells. Coelom may be formed by out pouching or splitting of mesoderm

3. Skeletal System:

Cartilage , Bones

4.Circulatory System:

Blood cells: RBCs , WBCs , Platelets

Blood vessels: Arteries , Veins , Capillaries , Heart

Q.70 Write Mechanism of Development: Nuclear equivalence?

1. Roux–Weismann Hypothesis:

- Nineteenth century
- How developing embryo develop multiple types of cells?
- Heredity material is distributed unequally
- Genome is broken down into smaller and smaller fragment

2. Hans Driesch Discovery:

1892

If two cell stage sea urchin embryo is mechanically shaken to get cells apart, both develop into normal larvae.Conclusion: both cells contain same genetic material

3. Hans Spemann Discovery:

Placed a minute ligature of hair around salamander zygote .Nucleus containing portion divides. Later at 16 cell stage, one nucleus move to anucleated portion and it also start cell division. When separated both developed normally

3. Hans Spemann

- Both halves developed normally only if both receive gray crescent
- This means cytoplasmic contents paly their role in formation of different types of cells

Q.71 Write Mechanism of Development : Cytoplasmic Specification?

1. Morphogenetic Determinants:

Cytoplasmic components that are unequally distributed within egg.These determinants control the development of particular cell type. Fate of blastomeres is specific by type of cytopalsm

2. Tunicate Example:

Some tunicates contain different colours in cytoplasm

- **Yellow:** muscle cells
- **Gray equatorial:** notochord and neural tube
- **Clear:** larval epidermis
- **Gray vegetal:** gut

3. Vertebrate Embryo:

Cytoplasmic specification is relatively less important .Gray crescent form dorsal lip of blastopore, is essential for gastrulation

4. Mosaic Development:

When a particular blastomere is isolated form embryo, it forms characteristic structure. In the absence of that cell the structure is missing. Mosaic development is characteristic of most protostomes

5. Regulative Development:

In many animals, fate of cell depends upon interaction with neighbors rather than cytoplasm itself.Each cell is able to produce entire embryo. If a blastomere is removed, the other change their fate to compensate this.

Q.72 Write mechanism of Development: Embryonic Induction?

1. Embryonic Induction:

The capacity of some cells to evoke developmental process in others. It is wide spread phenomenon in development

2. Spemann and Mangold Experiment:

- Embryonic induction was first reported

- The dorsal lip of blastopore when transplanted to later or ventral side of other embryo, it developed notochord, somite
- Also ectoderm of host
- Whole system formed

3. Primary Induction:

- Dorsal lip of blastopore can induce the development of whole embryo, it is called as primary organizer
- The phenomenon of induction by blastopore is called primary induction.

4. Secondary Induction:

- Formation of other type of cell by inductive process is termed as secondary induction

Q.73 Differentiate B/W Protostome Vs Deuterostome?

1. Protostomes:

Protostomes belong to group protostomia of animal kingdom

Together with deuterostomes,they are part of grade bilateria

Member phyla: Platyhelminthes, Rotifera, Nematoda, Annelida, arthropoda, Mollusca,

2. Deuterostomes:

Deuterstomes belong to group deuterostomia of animal kingdom

Together with protostomes,they are part of grade bilateria

Member phyla: Echinodermata, Hemichordata, Chordara

3. Cleavage Pattern:

Both differ in cleavage pattern.

Protostomes: Spiral cleavage as spindles fibers are oriented obliquely to animal vegetal axis

Deuterostomes: Radial cleavage as spindles fibers are oriented perpendicular to animal vegetal axis

4. Determinate and Indeterminate Cleavage:

Determinate: Fate of blastomeres is established at very early stage *i.e,* 2 or 4 cell stage. If some cells detach embryo fails to develop.

Indeterminate: Fate of blastomeres is established at later stages of development.

5.Development of Mouth and Anus:

Protostomes: The blastopore formed during gastrulation remain open and forms mouth while anus develops from opposite side of cavity.

Deuterostome: blastopore develops into anus while mouth develops from distal end.

6.Coelom Formation:

Protostomes: Coelom is formed by splitting of mesoderm. This pattern is called schizocoelous.

Deuterostome: Coelom is formed by out pouching of primitive gut (archenteron). This pattern is called enterocoelous.

7. Larval Stages:

Protostomes: Development of many protostome include a top shape larva called trochophore larva

Deuterostome: Some deuterostome have a Kidney-bean shaped larva called dipleurula.

Q.74 Define Metamorphosis and write its advantages and disadvantages?

Metamorphosis:

Metamorphosis is an abrupt change in structure and function from immature stage to adult. It takes place in insects, mollusks, crustaceans, cnidarians, echinoderms, tunicates, fishes and amphibians.

Instars:

During metamorphosis animals undergo a series of molts to grow, and each stage between molts is called an instar.Mosquito larvae have four instars. Butterfly larvae have five instars

Advantages of Metamorphosis:

- To complete development with food stage
- Reduced competition: Juvenile and adult stages have different mode of nutrition. Tadpole of frog is herbivore while adult is insectivore

Disadvantages of Metamorphosis:

- Short adult life while time spent on metamorphosis is pretty long

- Larva and adult do not share same food. This may be disadvantageous in food poor environment

Hormonal Control of Metamorphosis:

- **Vertebrates:** Thyroid hormones
- **Insects:** Prothoracicotropic hormone from insect brain
- This stimulates Ecdysone hormone released by prothoracic gland

Types:

- Complete metamorphosis
- Incomplete metamorphosis

Q.75 Define Holometabolous Metamorphosis and write its stages also?

Holometabolous Metamorphosis:

This is also called complete metamorphosis. In about 88% insects. It is the type of metamorphosis which separates the physiological process of growth from differentiation and sexual stages. Each process is completed without intervention of other stage

Four stages:

1. Egg **2.** Larva **3.** Pupa **4.**Adult

E.g: **Insects**

1. Egg Stage:

First stage of insect life cycle. Development starts after fertilization or sometime without fertilization. Early embryonic development completed at this stage. Egg stage is very short but in some cases undergo diapause

2.Larval Stage:

Second stage of holometabolous life cycle. Larvae are hatched out from eggs. Motile and worm like form. Actively feed and grow in size. Never possess wings. Simple eyes

3.Pupa Stage:

Third stage of holometabolous metamorphosis.Larva undergo metamorphosis to become pupa. Larva form a case or cocoon around itself. It is quiescent, non feeding developmental stage. Mostly pupa move very little, however some are mobile. Some insects undergo diapause as pupa. It is the stage of drastic changes in morphology and physiology

4.Adult Stage:

Final stage of metamorphosis. Adults have wings and reproductive organs. Most insects adult grow very little. Some do not feed and entirely focus on mating and reproduction

Q.76 Explain emimetabolous Metamorphosis?

Hemimetabolous Metamorphosis:

It is also called gradual incomplete metamorphosis.It is the mode of development of certain insects with three distinct stages

- Egg
- Nymph
- Adult

E.g., Aphid, grasshopper, cockroach

Egg Stage:

First stage of development.Initial stages of development are completed at egg stage.Egg stage varies in different groups but in some cases undergo diapause .Eggs of aphids are hatched before they are layed

Nymph Stage:

Nymph is an immature form hatched out from the eggs.Nymph lacks wings .They have underdeveloped reproductive organs.Nymph undergo gradual changes called instars.Aquatic insect may have tracheal gills

Adult Stage:

The apparent difference from nymph is presence of wings.They are also reproductively mature.

Q.77 Write hierarchy of Ecology?

Ecology:

It is the branch of biology that deals with the relationship of organisms to one another and with their physical environment

Ecological Hierarchy:

It is the way in which ecological components interact with each other

There are five levels

- Organism
- Population
- Community
- Ecosystem

- Biosphere

Organism:

Organism is the first level in ecological hierarchy.All the living things come at this level

E.g., human, lion, tiger, cow, tree, shrub .The way how organism interact with its environment is studied at this level

Population:

Group of organism of same species living at same geographical location at same time.At this level we study how the members of group live together and interact with each other

Community:

Groups of different species living together in a given area.The number of species represent the diversity of community.Community ecology include

- Predation
- Parasitism
- Competition

It is the study of all population in a community and their physical environment.Biotic components and abiotic components.How living organism make the use of abiotic components. How they get energy.

Q.78 Define ecosystem and write its Biotic and Abiotic Factors ?

Ecosystem:

It is the unit of ecology where living organisms interact with their physical environment

Ecosystems are very dynamic .They are controlled by different external and internal factors

Components of Ecosystem:

Ecosystem has two main components/factors

(i) Biotic factors/components
(ii) Abiotic factors/components

(i) Biotic Factors:

All living organisms in ecosystem make biotic factors.

Producers Consumers Decomposers

(ii)Abiotic Factors:

All nonliving things in ecosystem make abiotic factors.

Temperature Moisture Light

Soil Air water

Q.79 Write Abiotic Factors: Temperature , Moisture and soil?

Temperature:

It is the measure of heat present in a body. Animals extend a part of their energy on regulating temperature. Temperature influence metabolic rate

Heat Loss to Environment:

Conduction Convection

Radiations Evaporation

Heat Gain:

Radiations Conduction Metabolism

Effect on Habitat Requirement:

Thermoregulatory needs influence habitat requirement:

Availability of food Availability of water Availability of shelter

Body Size and Metabolism:

Larger animals have low metabolic rates. Smaller animals have higher metabolic rate. Mouse have 20 times higher metabolic rate than elephant.This is probably due to large surface to volume ratio for heat exchange

Abiotic Factors: Moisture

Moisture:

It is the amount of water diffused in environment as vapours, within solid or condensed on surface. In air the amount of water is also called humidity. High moisture….low water loss. Less moisture….more water loss

Gain of Water in Animals:

Animals acquire water from:

Food Drinking Metabolism

Loss of Water in Animals:

Water is lost along with

Urine Feces Gaseous exchanges Evaporative cooling

Measures to Reduce Water Loss:

Animals opt following strategies to reduce water loss. Dry feces.Concentrated urine or semisolid excretory wastes. Nocturnal behavior. Ability to store water

Abiotic Factors: Soil

Soil:

Soil reflects the geological and biological history .It effects the animals life directly and indirectly

Soil Characteristics:

Characters include

*Texture *Amount of organic matter *Fertility *Water holding ability

They directly affect the diversity of life on or in the soil

Soil as Habitat:

Soil is the habitat of large variety of animals including

Protozoa Nematodes Earthworms Arthropods

Salamanders Snakes Moles

Effect on Plant and Animal Life:

Soil have obvious effect on plant growing in that area. Different kinds of plants influence the animal life whether they are living in or on the soil

Q.80 Explain Population Growth?

Population Change Over Time:

Population are dynamic in nature.With the passage of time they are subjected to change due to:

- Birth
- Deaths
- Migration

Survivorship Curves:

Individuals within population are estimated by survivorship curve. It is the logarithmic plot between survivors and their age. Curves types are:

Type I Type II Type III

Type I Curve :

Also called convex population survivorship curve .Individuals of type I, population survive to an old age, then rapidly die. Environmental factors rarely cause mortality .Individuals live their potential life span .E.g. human population

Type II Curve :

Also called diagonal population survivorship curve .Individuals of type II, population have constant probability of death throughout their life.Environmental factors influence mortality

E.g., populations of birds and rodents

For Example, each cohort of birds experience 50% mortality each year.Potential life span of American robin is 11 years, however average life is 1.4 years

Type III Curve:

Also called concave population survivorship curve .Individuals of type III, population have very high juvenile mortality.Adults have much lower mortality rate.E.g., fishes and many invertebrates

Q.81 Explain exponential Growth?

Exponential Growth:

Population growth in which number of individuals increase by a constant multiple in each generation. Population increases by same ratio per unit time

Exponential Growth in Fruit Flies:

A few fruit flies in a large jar with abundant food.1 female lay 50 eggs.2 weeks for maturity.Each female of 2nd generation produce 50 more flies and so on

Biotic Potential:

The capacity of a population to increase maximally is called biotic potential.It is also called intrinsic rate of growth

Logistic Population Growth

Logistic Population Growth :

When a population reaches to carrying capacity of a particular environment, its growth is decreased. This results in logistic growth curve.Logistic growth curves reflect limited resources

Environmental Resistance:

The constrain of climate, food, space and other environmental factors is called environmental resistance.Due to these factors exponential growth can not occur indefinitely

Carrying Capacity:

The size of population that a particular environment can support is called its carrying capacity.It is symbolized by *K.*

Q.82 Write Population Density and animals inactivity?

Population Density:

It is the number of individuals living per unit area.Population density impact populations in predictable ways

Density-independent Factors:

These factors affect population regardless of their number

Weather conditions Earth quake Pollutants

Deforestation Hurricane Fire

Density-dependent Factors:

These are the factors which are more severe when population density is high

Competition for resources Disease Predation Parasitism

Animal Inactivity:

When food is scarce or environmental conditions are severe, the animals stop feeding and metabolic activities are decreased.

Torpor Hibernation Winter Sleep Aestivation

Torpor:

Period of decreased metabolism and lower body temperature in daily activity cycle

Hibernation:

Time of low body temperature and decreased metabolism from weeks to months.The set point of thermoregulatory center drop to 20°C

Winter Sleep:

Period of winter inactivity.Body temperature do not drop substantially.Animals can wake and become active. e.g., Bears

Interspecific Interaction

Q.83 Explain Competitive Exclusion and coexistance?

It states that two species competing for same requirement of food, habitat, nesting sites and other conditions of life can not coexist

Universality of Competitive Exclusion:

It influenced the scientists for more than 50 years.Numerous studies in nature failed to document its universality.This is because no two species have identical requirements

Example of Competitive Exclusion:

In California Golden-naveled scalesucker is a wasp parasite of red scale insect.Scientists introduced lingnan scalesucker from China .Within 2 years lingnan replaced golden-naveled scalesucker.

Coexistence:

It is the phenomenon in which competing species live together

How Coexistence Occurs:

- Coexistence can occur when species utilize resources in slightly different way
- MacArthur studied 5 species of warblers that use same caterpillar from different region of trees

Coexistence May Turn into Competition:

- By partitioning resource can be divided
- However, when resources are not enough then similarities become limiting factors and competitive exclusion may occur

Q.84 Write Coevolution and symbiosis?

Coevolution

It is a phenomenon in which ecologically related species sometimes coordinated such that each species exert a strong selective influence of the other

Role of Natural Selection:

Coevolution…..when competing for same resources or …… predator-prey relationships.Natural selection favours the development of selective characters.If balanced, no winner

Hummingbird and Passion Flower:

Ensifera ensifera is a hummingbird with extremely long beak.Passion flowers store nectar deep within tubular flower.Competition for nectar feeding is limited

Symbiosis:

Symbiosis is a close and long term association between two different organisms. Organism may get benefit or harm

Mutualism Commensalism Parasitism

Facultative and Obligate Symbionts:

Organisms are entirely dependent on symbiosis are obligate. *E.g.,* Fungus of lichen.Which can survive as independent and in association. *E.g.,* algae of lichen

Ectosymbiosis:

It is a symbiosis in which symbionts live on the body surface of the host or in the digestive tact.E.g., Lice, barnacles and cleaner fish

Endosymbiosis:

It is the relationship in which one symbiont live within the tissue of other..E.g., nitrogen fixing bacteria in root nodules, algae inside the corals

Q.85 Explain parasitism and predation?

Parasitism:

It is a relationship in which on organism (parasite) live in or on other organism (host). Parasite gets benefit while host is harmed.Host usually survives in this interaction

Ectoparasites and Endoparsites

Parasites living on the body surface of host are ectoparasites. E.g., lice, leeches.Parasites living within the body of host are endoparasites. E.g., tape worm, liver fluke, plasmodium

Intermediate Host and Definite Host

Some parasites have complex life histories with multiple host.Intermediate host: Immature stages and asexual reproduction of parasite.Definite host: host where parasite spends its mature life and reproduce sexually

Parasitoids

These are the animals that deposit eggs or spend developmental stages on the other animals. As development proceed they kill the host.They have properties of predator and parasite E.g., parasitoid wasps

Predation

It is type of interaction in which on animal kills and eat other animals. The animal which eats is called predator and the animal being eaten is called prey

Predator Prey Population

Interaction between predator and prey are complex .A large population of prey supports more number of predators.A decreased population of prey results in a decreased population of predators

Prey Distribution

Predators also effect the distribution of prey. For example, the lake trout of North America has never moved to freshwater of Siberian freshwater system.This is probably due to predation by Pacific sea lamprey.

Q.86 Write commensalism and camouglage and its types?

Commensalism:

It is the symbiotic relationship between two organisms in which one member benefited while other is neither helped nor harmed.

Remora Fish and Shark:

It is a small fish with disk on head.By these disks they get attached with large sharks and whales.When shark or whale feed they eat the scrapes of food.

Tree Frog and Plant :

Tree frogs live in green leaves of tree.They feed on insects and are protected from predators in green leaves of plants

Commensal May Become a Parasite:

A commensal do not harm its host. Sometimes environmental factor change and a commensal becomes parasite.E.g., *Entamoeba gingivalis* lives in mouth of human may cause teeth and gum decay occasionally.

Camouflage:

It is the phenomenon in which the colour pattern of animal helps to hide the animal from others

Two major types:

(i) Cryptic coloration
(ii) Countershading

(i)Cryptic Coloration:

- It is the type of camouflage in which animals takes pattern according to surroundings.
- Some prey animals are camouflage from predators

- E.g., arctic hare hare white fur coat in winter and brown in spring
- Predators use camouflage to avoid detection by prey
- Prey use camouflage to avoid detection by predators
- Asian tiger's rufous-red coat with black strips help it to undetected during stalking

(ii)Countershading:

- It is a camouflage common in eggs of frog and toad
- Darkly pigmented on top while light pigmented on the bottom
- From top dark top mixes with dark bottom
- From bottom light colour blends with light air water interface

Q.87 Write aposematic Coloration and mimicry?

Aposematic Coloration:

Phenomenon in which animals protect themselves by advertising themselves dangerous or distasteful.

They actually warn the predators to not attack on them

Dandrobatid Frogs of America:

Dandrobatid frogs of central and south America have poisonous secretion and are brightly colored.Their bright colour is a warning to the predators

Mimicry:

It is the phenomenon in which one species gains resemblance with other species and gets protection from predators.This resemblance may be with one or more than one species

Batesian Mimicry:

When one species resembles with other species and protected by aposematic coloration.
Mimic and model. Flies belonging to family Syrphidae resemble with bees or wasps, thus protected.

Mulerian Mimicry:

When mimic and model both are distasteful.If predator eats any one of these, It will not prey any one of them.

Q.88 Write Community Structure?

Community Structure:

Communities are not the random mixture.They are organized entities.Some members of community have greater importance in organizing community

Dominant Species:

Species that is responsible for determining community characteristics.E.g., a stream having a large population of rainbow trout that will determine the population of certain invertebrates

Keystone Species:

These are species on which several other species are dependent.Removal of keystone species will cause the death of dependents

Temporal Structure:

Cyclic activities also characterize the community.Community structure based on cyclic activities of daily or seasonal basis is called temporal structure of community

Community Diversity:

Variety of animals present in a community.It is also called community richness. Stable climate, more resources, high productivity, moderate predation promote diversity.

Q.89 Write ecological Niche , food chain and food web?

Ecological Niche:

It is the role of an organism in an ecosystem.It include all the attributes of life style of an animal.Where it lives.What it eats.Where it search for food. Physical parameters it prefers

Niches Overlap:

Ecological niches of two species may overlap.This will result a competition between them.Animals will compete for same resources

Niche Diversification:

Competition between species may be reduced if resources are divided .One part of resources by one species and second part by other species.This mechanism is called niche diversification

Food Chain:

Food chain:

It is the sequence of organisms in an ecosystem where energy moves from one organism to the next organism. It tells us the sequence to eat and being eaten

Grazing Food Chain:

It is the food chain that starts from the living green plants.Green plants are eaten by herbivores which are eaten by carnivores.Such food chains are directly dependent on sunlight

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Detritus Food Chain:

It is the food chain that starts from dead organic matter to microorganisms and then to organisms.They are little or not at all dependent on direct solar energy

Importance of Food Chain:

To understand the feeding relationships between organisms

To explore the energy flow mechanisms and matte circulation

To understand the problems of biomagnification.

Food Web:

Food web:

It is a complex network of organisms through which energy and matter is transferred in an ecosystem Food web consists of many interconnected food chains

Grazing Food Web:

These are the food webs where plants make the starting point. Food webs based on algae and green plants

Detrital Food Web:

These are the food webs where decomposers make the starting point. Bacteria and fungi move the food material to the carnivores

Trophic Levels:

As food webs are very complex.The animals feeding in a food web are distributed into different trophic levels.Autotrophs make 1st trophic level.

Q.90 Write different Trophic Structure?

Producers

Producers:

They obtain nutrition from inorganic compounds.Also called autotrophs.Form the first trophic level of an ecosystem

Two types:

- Photoautotrophs
- Chemolithotrophs

They use sunlight as energy source for the synthesis of food from inorganic compounds.Green plants, algae, and photosynthetic bacteria.Photosynthesis transform less than 1% of sunlight reaching to earth.30% reflects back .20% is absorbed by atmosphere.50% absorbed by earth

Chemolithotrophs:

They obtain energy by oxidizing inorganic compounds.Usually live in extreme environments

E.g., sulfur bacteria

Consumers

Consumers:

Organisms which can not synthesize their food from inorganic compounds.They eat other organisms to get nutrition.All animals are consumers

Primary Consumers:

They are also called herbivores.These are the animals which feed on plants

E.g., insects

grain eating birds .reptiles, birds and mammals which feed on leaves and other vegetation

Secondary Consumers:

Also called carnivores.These are the animals which feed on herbivores.E.g., lion, tiger, frog

Tertiary Consumers:

These are the animals which feed on secondary consumers.E.g., a snake when eating a frog, a jaguar when hunting for a crocodile

Decomposers

Decomposers:

These are the organisms which obtain energy by breaking down dead organic matter into simpler compounds.Also called saprotrophs

Bacteria and Fungi as Decomposers:

Chief decomposers of an ecosystem are bacteria and fungi.They release extracellular enzymes which breakdown substrate molecules.Organisms at all trophic level when die are used by decomposers.

Recycling of Nutrients:

If there are no decomposers mounts of dead bodies will be every where.All useful minerals will remain lock in dead bodies.There will be little space to live.Decomposition recycle the nutrients in ecosystem

Q.91 Ecological Efficiencies and ecological pyramids?

Ecological Efficiencies:

It is the efficiency by which animal at a trophic level convert food into new biomass.A large amount of food is not digested by the animals

Efficiencies of Animals:

Herbivores have lot of plant biomass, but they are less efficient.Carnivores are more efficient.At an average 10% biomass conversion efficiency.Efficiencies range for less than 1% for most herbivores to more than 35% for ectotherm carnivores

Ecological Pyramids

Ecological Pyramids:

It is the graphical representation of relationship between organisms at different trophic levels of an ecosystem.Three most common types of pyramids:

(i) Pyramid of number
(ii) Pyramid of biomass
(iii) Pyramid of energy

(i)Pyramid of Numbers:

It is the graphical representation of total number of different organisms present at each trophic level.This shows the population of organisms in a food chain

(ii)Pyramid of Biomass:

It is the representation of total mass of living matter present at each trophic level.Biomass decrease at each ascending trophic level .Biomass is measured in grams of dry mass per square meter

(ii)Pyramid of Energy:

It is the representation of total amount of energy present at each trophic level in an ecosystIt is always upright .Energy pyramids help to compare the productivity of ecosystems.

Q.92 Write Ecological Pyramids: Examples?

Pyramid of Numbers Example:

In a grass land the survey was conducted to check the trophic structure.

Following data was obtained

- Producers: 1.5×10^6
- Primary consumers: 2×10^5
- Secondary consumers: 9×10^4
- Tertiary consumers: 1

Pyramid of Biomass Example:

In a pond the survey was conducted to check the trophic structure.

Following data was obtained

- Producers: 809 g/m^2
- Primary consumers: 37 g/m^2
- Secondary consumers: 11 g/m^2
- Tertiary consumers: 1.5 g/m^2

Q.93 Explain Energy and Matter Flow in an Ecosystem?

1. Energy Budget:

It is the balance sheet of energy gained and energy expenditure of an organism.An organism is an open thermodynamic system which exchange energy with its surrounding in three ways i.e., heat, work and potential energy of biomolecules

Energy budget allocation varies for ectotherms and endotherms.Ectotherms rely on environment as heat source.Endotherms rely on metabolic heat to maintain body temperature

2.Energy Flow:

Organisms require energy in one form or the other.Energy flow in an ecosystem is major factor for survival for great number of organisms.Source of energy is sunlight.Energy flows via food chain and food web.Flow is by 10% law.Energy flow is unidirectional

All elements essential for life are derived from environment.They make all form of molecules for life.When animals die and their dead bodies are decomposed, these materials are recycled in the ecosystem.Nutrient flow in ecosystem in a cyclic way.

Q.94 Explain Carbon Cycle ?

Importance of Carbon:

Carbon is the basic element of all organic molecules.All biomolecules have a carbon backbone.It has ability to bond with four other atoms at the same time

Carbon Reservoir:

It is plentiful on earth and is rarely limiting factor.Reservoir for carbon is CO_2 in atmosphere and water.When organic matter is completely oxidized, it releases CO_2 into atmosphere

Carbon Source in Aquatic Environment:

In aquatic environment, CO_2 combines with water to form H_2CO_3.When CO_2 level in water decreases H_2CO_3 breaks to release CO_2

Carbon in Calcium Carbonate:

Some carbon is tied up in the form of calcium carbonate ($CaCO_3$).This $CaCO_3$ is in the part exoskeleton and endoskeleton of mollucs, echinoderms, and other invertebrates .They ultimately form limestone reservoirs

Carbon in fossil fuels:

A supply of carbon is also tied up in fossil fuels.During 65 million year of Carboniferous period (280mya), organic matter got buried and converted into coal, oil and gas.When fossil fuels are used, carbon returns to atmosphere.

Q.95 Explain Nitrogen Cycle?

Importance of Nitrogen:

Nitrogen is an essential component of proteins, nucleic acids and ATP.It is also found in membrane lipids and modified carbohydrates

Reservoir of Nitrogen:

Atmosphere is main reservoir.About 78% nitrogen present in air.But this is not directly available to plants.Plants can use nitrogen in the form of NH_3 and NO_3^- .Animals obtain N from plants.

Nitrogen Fixation:

Gaseous nitrogen converted to NH_3.Aqueous form of NH_3 is NH_4^+.Bacteria mainly in the roots of leguminous plants and cyanobacteria do this job.

Lightening , Industry

Nitrification:

It is the process by which NH_3 is converted to nitrate (NO_3^-).It is a two step process.In first step NH_3 is converted to nitrite (NO_2^-) by Nitrococcus bacteria.In second step NO_2^- is converted to NO_3^- by Nitrobacter.

Ammonification:

The process by which organic nitrogen is converted into ammonia is called ammonification.Ammonia released in this process may be directly assimilated by the plants or used after nitrification

Denitrification:

The process by which nitrates are converted back to nitrogen ,This process occurs is completed by anaerobic bacteria. e.g., Pseudomonas, Aachromobacter

Q.96 Explain Phosphorus Cycle?

Importance of Phosphorus:

Essential component of:

DNA RNA ATP Phospholipids

Vertebrate skeleton Teeth

Reservoir of Phosphorus:

Sedimentary rocks contains deposits of phosphorus.Extraction of phosphorus is done by weathering.Rain and other erosion processes bring phosphorus back to soil

Absorption by Plants and Animals:

Plant, microbes and fungi absorb phosphorus from soil.Animals using phosphorus by utilizing plants. Some phosphorus also come into through water.Plants and animals die, phosphorus is return back to the environment via decomposition.Living organisms may use this phosphorus again.Some phosphorus may run along with water and become part of sediments

Q.97 Explain Sulfur Cycle?

Importance of Phosphorus:

Essential component of:

Proteins Enzymes Vitamins

Reservoir of Phosphorus:

Abundant in nature. Bright yellow crystalline solid. Most of it is tied up in rock and buried in depth of oceans. Atmosphere also contain oxides of sulphur

Mineralization:

It is the process which organic sulfur is converted into inorganic form. Hydrogen sulfide, elemental sulfur and sulfide minerals are formed

Oxidation:

At this step sulfides and elemental sulfur are converted to sulfates . Sulfates are reduced by plants, fungi and bacteria and incorporated into organic compound.

Q.98 Explain Water Cycle?

Water on land:

Essential for life. Life can not exist without water. 97% water is marine with high salt concentration

3% freshwater, mostly glaciers and groundwater . Less than 1% of water is available to human

Evaporation:

Water in the oceans, lakes, swamps, rivers and soil when heated is turned into vapors.Similarly water also evaporates from surfaces of plants and animals. Transpiration. Sublimation

Precipitation:

Water vapors move up higher, condense to form clouds.Clouds are pushed around the world by moving air.Further cooling causes precipitation.Precipitation may be in the form of rain, hail or snow.78% occur over oceans

Runoff:

It is the flow of water across the land

It includes

- Surface runoff
- Channel runoff

Under ground water moves slowly and may remain in aquifer for thousands of years

Eventually is reaches to the oceans.

Q.99 Define Biosphere and write its Subdivision?

Biosphere:

Thin outer layer of earth which can support life.It includes all life on earth and the physical environment on which living organisms exist and interact

Subdivision of Biosphere:

Life exists in soil, water and air.Biosphere is subdivided accordingly

(i) Lithosphere
(ii) Hydrosphere
(iii) Atmosphere

(i)Lithosphere:

It is the rigid material on the surface of earth.Most of the terrestrial animals live on lithosphere.Lithosphere is the ultimate source of all mineral elements required by all living things

(ii)Hydrosphere:

It is the water on the surface of earth. It extends into the lithosphere and atmosphere. Water is distributed on earth by global hydrological system. Water returns to oceans by precipitation

(iii)Atmosphere:

It the gaseous content of biosphere. Atmosphere extends up to 3500 km above the surface of earth. Life in atmosphere is present in lower 8-15km called trophosphere. The main gases in troposphere are nitrogen, oxygen and carbon dioxide.

Q.100 Explain Ecosystem and Biome?

Ecosystem:

It is the interaction between living and non living components of the environment

Biome:

Biome is an area of earth with distinctive association of plants and animals

Major biomes of the world include:

Tundra Coniferous forests Deciduous forests

Grasslands Deserts

Biomes are Identified on the Basis of Plants:

Botanists divided terrestrial environment long ago into different biomes.Characterization of biome on the basis of animal life is difficult.A biome is identified by dominant plant formation.Because plants support a particular type of animals, so the biomes also have characteristic type of animals

Ecocline:

The boundaries of biome are not distinct.It contains tree of both types of biomes.This indistinctive boundary having a mixture of plants is called ecocline

Cell:

Air circulation in each hemisphere is broken into three zones from poles

1. Zero to 30° latitude
2. 30° to 60°
3. When cold air move back to poles at high altitude.

Q.101 Define Tundra Ecosystem and also write its characteristics and types?

Tundra:

Treeless region found in arctic and top of mountains with severe cold climate and permanently frosted subsoil

Characteristics of Tundra:

Climate: a long winter is interrupted by a short summer. Plant can grow in this summer of 60 days

Soil: remain frozen in most of the year. Subsoil permafrost.Most tundra region is covered with bogs, marshes, ponds and decayed vegetation

Average precipitation: is very low, less than 25cm/year

Temperature: cold environment, average annual temperature is -10°C although it may drop

to -50°C in winter and rise to 10°C in summer

Plant life: no trees, lichens, grasses and dwarf woody plants. Growth of plants is very slow due to extreme conditions

Animal life: Rodents such as arctic hare, musk oxen, arctic foxes, owls, weasels and caribou (reindeer)

Types of Tundra

Tundra exists in two forms:

(i) Arctic tundra
(ii) Alpine tundra

(i)Arctic Tundra:

It is present in northern hemisphere. It includes the areas of northern Russia and Canada. It is the home of Nganasan and Nenets people. Temperature ranges form -30°C to 20°C. Highly vulnerable environment. Animal life include arctic fox, polar bear, gray wolves, caribou, snow geese and musk ox . Plant life lichen, mosses, grasses and shrubs

(ii)Alpine Tundra:

High elevation are too cold and windy to sport plant life. North American Cordillera, the Alps and Pyrenees of Europe, the Himalaya and Karakoram of Asia, the Andes of South America, and the Eastern Rift mountains of Africa

Climate Change Impact on Tundra:

Arctic tundra changing dramatically due to global warming.Animal previously not found in arctic tundra are now competing with native animals.Shrubs are getting taller and animal like spider are more active.Permafrost is melting .It holds large amount of carbon and methane.Climate change may change tundra from carbon sink to carbon contributor.

Q.102 Define Coniferous Forests and write its characteristics and its location?

Coniferous Forests:
Coniferous forest are dominated by cone bearing, needle shape evergreen plants called conifers

Characteristics of Coniferous Forest:

Climate: cool summer and cold winder, growing season is 130 days

Soil: soil is acidic and relatively infertile

Average Precipitation: moderate rainfall, 40-100cm/year. Most precipitation in the form of heavy snow

Temperature: average annual temperature is 10°C although it may drop to -40°C in winter and rise to 20°C in summer

Plant life: Spruce, fir and pine. They are evergreen. Some primary production also on warmer summer days

Animal life: low diversity but large population. Snowshoe hare, lynx, wolves, caribou, moose and summer moths

Primary Productivity

It is the main feature of an ecosystem.Coniferous forests are evergreen, their primary productivity is higher than grasslands but lower than deciduous and tropical forests

Coniferous Forests Location:

It is the largest terrestrial biome on earth

Taiga: Stretched from North America to Europe and Asia in northern hemisphere

Temperate Coniferous Forests: They are present worldwide at high altitudes in America, Africa, Europe South Asia

In Pakistan they are present at 1000 to 4000 altitude

Chitral , Swat , Dir , Abbot Abad

Hilly Areas of Balochistan , Murree , Azad Kashmir

Human Impact:

Deforestation: Coniferous tree wood is used in buildings and furniture

Climatic changes:

- Mild and short winter
- Acid rain

Q.103 Explain Temperate Deciduous Forests?

Temperate Deciduous Forests:

These are forests with moderate climate and well define summer and winter.They drop their leaves during autumn

Climate:

Four distinct seasons. Winter, spring, summer and fall. Annual precipitation ranges between 75 to 150cm/year. Temperature varies from -30°C to 30°C

Soil Conditions:

Soil is fertile. It is brown in colour. Decomposition of fallen leaves enrich the soil. Primary productivity is higher than grasslands and coniferous forests

Plant Life: Broad leaves trees

Animal Life: Great diversity of animal life

Location:

Eastern USA , Canada , Japan

China , Some part of Australia

Human Impact:

- Valuable timber
- Mostly cut down from Europe for farming purposes

- Some have been converted to commercial forestry.

Q.104 Write the Grassland its type and location ?

The Grassland:

These are the areas where vegetation is mainly dominated by the grasses.They account for 20–40% of world land area.They lie between forest and deserts

Climate:

Grassland is found where there is not enough rain fall.Precipitation is 25 to 100cm annually.Temperature ranges from -20°C to 30°C

Soil Conditions:

Soil is porous.The upper layer is nutrient rich making it ideal for agricultural use.But the low water availability may eventually convert them into deserts

Types of Grasslands:

Grasslands of North America are called **prairies**.Grasslands of South America and Africa are called tropical **savanna**

Plant Life:

Grasslands are dominated by different types of grasses. Grasses make layering.Prairies are normally without trees while savanna has scattered trees (less than 30%).Clover, salvia, wheat, barley, coneflowers. many species of grasses including, purple needle grass, wild oats, foxtail, ryegrass, and buffalo grass

Animal Life:

Grasslands support large populations of herbivores and carnivores ,Insects , Deer , Zebra

Wildebeest , Giraffe , Bison , Lions, tigers, hyenas

Location:

Great plains of North America. Pampas of south America. Veldt of south Africa. Steppes of Central Eurasia. Surrounding the desert of Australia

Human Impact:

- Continuous global warming
- Conversion to agricultural land
- Urban development

Q.105 Write the Chaparral and its location?

The Chaparral:

It is the biome fount in small patches in most of the continents.It is characterized as very hot and dry

Climate:

Winter is very mild and usually about 10°C. Summer is hot and dry with temperature up to 37.5^{o}C. Most of the rain occur in winter. Annual precipitation is 30-75cm

Soil Conditions:

Soil is thin and rocky. Nutrient poor. Highly susceptible to erosion. Soil contain low amount of moisture

Plant Life:

Plants of chaparral are low and shrubby.They have tough and waxy leaves.They are broad leaves and evergreen.Plants have no thick bark so not very resistant.But recovery is very fast

Types of Plants:

Two types of plants

Sprouters: dormant meristem, storage and intact root help to regenerate quickly

Seeders: long lived seeds, germination induced by fire .Plants include toyon, chamise, poison oak, scrub oak, Yucca, Sumac

Animal Life:

Animals include Insects Rodents Rabbits

Lizards Snakes Mule deer

Location:

West coast of United States and South America…South Africa.Western tip of Australia..Coastal Area of Mediterranianean.

Q.106 Explain Desert Ecosystem?

Desert Ecosystem:

A region which receives annual precipitation less than 25cm.They are characterized by scarce vegetation and extreme temperature occilation

Climate:

Extremely dry climatic conditions.Lowest precipitation in comparison with other biomes.There go years and years without precipitation.At day time temperature may be 38 while at night it may be 4°C

Soil Conditions:

Desert soil is dry. Low organic nutrients

Low fertility . Low water holding capacity

Plant Life:

Only those plants can grow which ability to survive in extreme conditions of drought.Reduced leaves, succulent stems, deep stomata and waxy cuticle are the important adaptations of desert plants

Animals life:

Animals of desert include Insects, Other arthropods.

Reptiles: snakes, lizards

Birds: road runner, cactus wren, turkey vulture, burrowing owl

Mammals: mule deer, jackrabbit, kangaroo rat, ground squirrel

Deserts of the World:

Antarctic desert , Arctic desert , Sahara desert , Australian desert

Arabian desert , Gobi desert , Kalahari desert , Thar desert

Columbia basin , Dasht-e-kavir.

Q.107 Write Tropical Rainforests and its location and impact on human?

The tropical rainforests:

It is a hot moist biome found near the equator. They are the richest and most exciting areas of the earth

Climate:

Warm temperature and abundant rainfall.Average temperature ranges from 20°C to 25°C. It remains constant.Annual rainfall is more than 200cm.Little seasonal variation in daily length.

Plant life:

Very diverse plant life.Thousands of species .Non of them dominating.A single hectare typically contains 50 to 70 species.Forests are highly stratified.Trees are broad leaved and non–deciduous

Plant Types:

50% of plants and animals with 6% land. Plants grow faster. Bigger leaves. Plants include

Palm species. Rubber tree. Bamboo. Orchid. Tualang. Different epiphytes. Pitcher plant

Animal Life:

Stable temperature and high humidity is ideal for both soft bodied and hard bodied endotherms

Annelids , Arthropods , Amphibians , Reptiles , Birds , Mammals

Geographical Location:

South and central America , Western Africa , Southeast Asia , Australia

Human Impact:

Deforestation: once they were 14% and now they are 6%. It is estimated that in next 40 years the remaining will be lost as well

Habitat loss , Extinction , Soil erosion , Climate change

Q.108 Explain Freshwater Ecosystems: Lotic Ecosystem and lentic ecosystem?

Lotic Ecosystem:

The Lotic Ecosystem:

It is the aquatic ecosystem of flowing water. It includes

Brooks , Streams , Rivers

Oxygen Supply:

Lotic systems typically have high oxygen level.Turbulence of water while flowing over rocks, logs and other objects causes oxygenation

Plants and Animal Life:

Mainly algae and attached aquatic plants are indigenous producers.Animals include different aquatic invertebrates, larvae and insects.Fishes, amphibians reptiles birds and mammals are adapted to lotic habitat

Effects of Water Currents:

It is the most important physical factor of lotic habitat. Animal adaptation include. Dorsoventral surface. Strong swimmers. Burrowing . Attachment with substrate

Lentic Ecosystem:

The Lentic Ecosystem:

This is the ecosystem of standing water. Lentic ecosystem include:

Lakes , Ponds

Vertical Stratification:

Vertical stratification is common especially in temperate region.In summer, upper layer is warm while lower layer is cold and more dense, so no mixing .In fall, upper layer is dense so mixing of water occurs.Phytoplankton are main producers of lentic ecosystem

Oxygen Supply:

Deeper layers are depleted of oxygen in summer.In fall the mixing of entire water column supply oxygen to deeper layers

Oligotrophic and Eutrophic bodies:

Deep, cold, nutrient poor lakes are relatively unproductive called as oligotrophic .Shallow warm, nutrient rich lakes are highly productive called as eutrophic

Plant and Animal Life:

Plant life include phytoplankton, periphyton algae, emergents, floating-leaves, submersed and free floating.Animal life include zooplankton; invertebrates like crabs, crayfish and shrimps; molluscs.

Q.109 Explain Marine Ecosystems the Estuaries and Intertidal Zone?

The Estuaries:

Estuaries occur where freshwater meets seawater.This dilutes the salinity of seawater

Productivity of Estuaries:

Estuaries are highly productive because

I. Streams carry nutrients of freshwater

II. Tides bring nutrients from ocean

Plant and Animal Life:

Producers include phytoplankton, attached aquatic vegetation and emergent vegetation such as cordgrass.Estuaries are nurseries for crustaceans, molluscs and marine fishes. Feeding grounds for wading and swimming birds

Human Impact:

Human have drained and filled estuaries to subdivision.Polluted by oil exploration and drilling.Acids rain have threaten the productivity

Marine Ecosystems: Intertidal Zone

The Intertidal Zone:

Extend from splash zone ocean to low tide marks.This region is subjected to the action of both tides and waves

Environmental Fluctuations:

It is the harshest and the richest of all marine environments.Here is pounding surf, sun, wind, rain, erosion, sedimentation and temperature fluction

Plant and Animal Life:

There is bounty of habitat and plenty of nutrients.Plants are larger. They include seagrass, red algae, mangroves.Motile animals like fishes and crabs move with tides.Other such as amphipods, isopods, ghost crabs and polychaetes burrow in the sand and mud.

Q.110 Explain marine ecosystem: Neritic , Oceanic and Coral Reefs

Marine Ecosystems: Neritic Ecosystem

The Neritic Ecosystem:

Neritic ecosystem extend from littoral zone to the edge of continental shelf.It is also called the shallow water

Productivity:

Nutrients are delivered by rivers and upwelling at the edge of continental shelf.This region is more productive than open sea.Planktonic and attached algae contribute to the primary productivity

Plant and Animal Life:

Plants include kelps, algae, seaweeds, seagrasses and phytoplankton.Invertebrate animals include crabs, sponges, echinoderms, corals, and polychaetes

Marine Ecosystems: Oceanic Ecosystem

The Oceanic Ecosystem:

It starts from continental shelf.It is the open sea

Productivity:

It is relatively unproductive.Virtually all primary productivity is in the upper water

Plant and Animal Life:

Phytoplankton in the upper water are the main producers of oceanic system.Invertebrate animals include worms, echinoderms, molluscs.Vertebrates include fishes from small fishes to whales and sharks.At depth no light so the animals here have luminescent organs which serve as torch light.These animals depend on nutrient coming from upper layers

Marine Ecosystems: Coral Reefs

The Coral Reefs:

It is the underwater ecosystem.They are formed by colonies of corals. Polyps are held together by calcium carbonate.They are the association of stony coral and algae.Coral are members of class Anthozoa.Algae are zooxanthellae

Occurrence:

Reefs are formed over thousands of years in warm, clear and relatively shallow waters.The greatest depth for their growth is 50-60 m.Below this light is insufficient to support zooxanthellae

Rich Biodiversity:

They are the richest source of biodiversity in oceans.Coral reefs occupy less than 0.1% of oceans, yet they are home of more than 25% marine species.Residents include sponges, cnidarians, nematodes, crustaceans, molluscs, echinoderms, tunicates and fishes.

Q.111 Explain Ecological problems?

(i) Human Population Growth

Population Growth:

It is the increase in number of individuals in a population.Expanding population is the root cause of most environmental problems.More population means more need for resource: food, urbanization, industrialization

Earth's Carrying Capacity:

How many human can be supported by earth?,Virtually all scientists agree that even 5 billion is more for earth to support with all facilities.Maximum carrying capacity of earth is may be 9 to 10 billion people

Age Structure:

Age structure shows the proportion of individuals at prereproductive, reproductive and postreproductive stage.Developing countries have more infants means high birthrates.In contrast to human, in natural population high birthrate is accompanied with high mortality

(ii) Pollution

Pollution:

Pollution is any detrimental change in ecosystem.Mostly is results from human activities.More human population<more demands<pollution

Water Pollution:

Industries generate toxic waste. Sewage waste. Agricultural runoff

Air Pollution:

Burning of fossil fuels release CO_2, oxides of sulfur and nitrogen.Acid deposition by these oxides.Increased CO_2 causes green house effect.Global warming

Land Pollution:

Oil spillage , Plastics , Pesticides , Toxic industrial waste

Garbage , Buildings leftover

Biological Magnification:

Wastes and poisons enter in food chain and travel to the organisms at highest level of food chain including human.Thy are not biodegradable and interrupt physiological processes

(iii) Resource Depletion

Natural Resources:

Materials or substances occurring in nature which can be exploited for human benefits

Important are:

Fossil fuels , Minerals , Air , Water , Soil , Biodiversity

Depletion of Resources:

More population , Better life , Overexploitation , Fossil fuels , Minerals

Air , Water , Land/soil , Biodiversity

(iv) Biodiversity Loss

Biodiversity of Earth:

More than 1.6 million

Estimate is 4 to 6 million

More accurate is between 8.7 ± 1.3million

Threats to Biodiversity:

Biodiversity of all natural areas is threatened

Acid rain , Pollution , Urbanization , Agriculture

Human is exploiting 40% of earth net primary production. Habitat loss

Solutions:

Realization about biodiversity. 22nd May is World Biodiversity Day . A system of international ethics that value natural diversity.

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